# Study Project: Adversarial Machine Learning

Asya Mitseva, M.Sc.
Prof. Dr.-Ing. Andriy Panchenko

Chair of IT Security
Brandenburg University of Technology Cottbus-Senftenberg

15 October 2025

Brandenburg
University of Technology
Cottbus - Senftenberg

# General Information (1/3)

- **Lecturer:** *Prof. Dr.-Ing. Andriy Panchenko*
- **Coordinating teaching assistant:** *Asya Mitseva, M.Sc.*
- **Contact:** asya.mitseva@b-tu.de
- **Online Platform:** *Moodle*
  - Available slots for consultations and Q&A sessions
  - Announcements, distribution of exercise sheets, etc.
- **ECTS Credits: 8**
  - Study project: 165 *hours* $\approx$ 15 *hours per week per person*
  - Self organized studies: 60 *hours per person*
  - Consultation: 1 hour per week per person
- **Assessment**
  - Graded Q&A session for each exercise sheet
  - Final presentation of your study project
  - Final report

# General Information (2/3)

- **Master students eligible to register for this course**
  - Cyber Security Master
  - Computer Science Master
- Students enrolled in other study programs are **not eligible** to register
  - eBusiness Master
  - Artificial Intelligence Master
  - Künstliche Intelligenz Technologie Master
- **The course is not suitable for first semester Master students**
- **Restrictions for the creation of groups**
  - Groups of two students, repeating the course
  - Groups of two students, taking the course for the first time
  - *Groups of one repeater and one non-repeater currently not possible*

# General Information (3/3)

- **New:** The course has a **limited** number of **participants**
  - Registration deadline has already passed (12 October 2025)
  - Only officially registered students will be considered during the selection
  - The test is only one of the selection criteria
- **Advice:** If you do not get place, do **not deregister** immediately
  - If selected students drop the course, we continue filling the free spots
  - The process continues until the release of the first practical sheet
- **Request:** Let us know if you decide to drop the course
  - Via email at asya.mitseva@b-tu.de
  - Then, other students get the chance to take the course
- Final deregistration of not selected students is handled by us
  - The deregistration is *not counted* as a failed attempt!

# Organization of Study Project (1/4)

- **Lecture period:** from 13.10.2025 to 06.02.2026
  - All practical sheets and final presentations should be done in this period
  - No extensions of deadlines possible
- **Working groups:** two students in a single group
- **Exercise sheets**
  - Five practical exercise sheets
  - *Biweekly submissions and Q&A sessions* of practical exercise sheets
    - *Continuous assessment: failed submission/Q&A session ≡ failed course*
  - Final (sixth) exercise sheet for preparing final presentation and report
- **Consultations and Q&A sessions**
  - Weekly announcement of available time-slots for consultation
    - *General rule: First come, first served!*
  - *Avoid any (last-minute) requests for consultations via email!*
  - Predefined time-slot for each group for *Q&A session*
    - *Always take place on Wednesday*

# Organization of Study Project (2/4)

- **Deadlines**
  - Cancellation of course registration possible *before 02.11.2025*
  - Submission of exercise sheet: *biweekly, by Wednesday, 11:59 a.m.*
  - Release of new exercise sheet: *biweekly, Wednesday at 12:00 p.m.*
- **Grading system**
  - Number of *points* for a *single practical sheet*: *maximum 12*
  - Number of *points* for the *final presentation*: *maximum 10*
  - Number of *points* for the *final report*: *maximum 10*
  - *At least 50% of the corresponding points* needed *to pass* assignment
  - *What if you did not get at least 50%?*
    - In total, you have *two chances for resubmitting* during the whole course
    - *Otherwise, you fail the course*
  - *At least 75% of the total number of points for practical sheets* to be eligible to do final presentation; *otherwise, you fail the course*
  - *At least 70% of the total number of points* for all practical sheets, report, and final presentation to *pass the course*

# Organization of Study Project (3/4)

- **Overall overview**

15.10.2025 – Kick-off meeting

16.10.2025 – *Announcing the list of selected students*

22.10.2025 – *Assignments of topics and release of first practical exercise sheet*

29.10.2025 – Consultation period

05.11.2025 – Consultation period

12.11.2025 – *Release of second practical exercise sheet and Q&A sessions*

19.11.2025 – Consultation period

26.11.2025 – *Release of third practical exercise sheet and Q&A sessions*

03.12.2025 – Consultation period

10.12.2025 – *Release of fourth practical exercise sheet and Q&A sessions*

17.12.2025 – Consultation period

07.01.2026 – *Release of fifth practical exercise sheet and Q&A sessions*

14.01.2026 – Consultation period

21.01.2026 – *Release of sixth exercise sheet and Q&A sessions*

28.01.2026 – Consultation period

02.02.2026 – 06.02.2026 – *Final presentations*

# Organization of Study Project (4/4)

- **Final presentation and final report**
  - Duration of the presentation: approx. 30 min (excl. project demo)
  - Use of *our LaTeX template* for the report is *mandatory*
  - General hints
    - Give clear and concrete problem definition of your project
    - Briefly motivate why the problem is interesting scientific problem
    - Point out the challenges that have to be addressed to solve the problem
    - Provide complete and convincing explanation how you solved the problem and argue why you chose the techniques presented
  - Other additional information will be announced during the semester

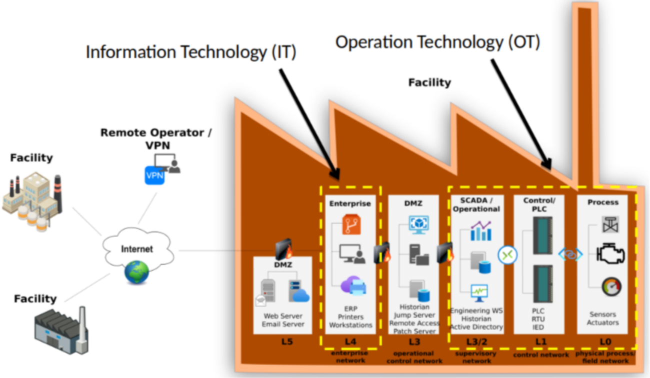
- **Last remarks**
  - Try to understand your problem first before start writing source code
  - Try to provide complete and precise solutions
  - Do not try to copy-paste old solutions from previous years
    - We have them as well ☺

# Some of Our Topics for Study Project

- **Overall idea**
  - Mount different attacks and/or defenses
  - Using machine learning approaches
- **Some of the possible topics**
  - Attack Detection in Industrial Control Systems (ICS)
  - Generation of Attacks for Industrial Control Systems
  - Implementation and Evaluation of Honeypots
  - Generation of Datasets for Anomaly Detection
  - End-to-end Traffic Correlation Attacks
- **Concrete topics may still vary and number of possible topics might be extended**

- **Industrial Control System (ICS)**
  - *Objective:* Monitor and control various industrial processes
  - *Examples:* Chemical plants, power systems, gas pipelines, etc.

- **General architecture of ICS**

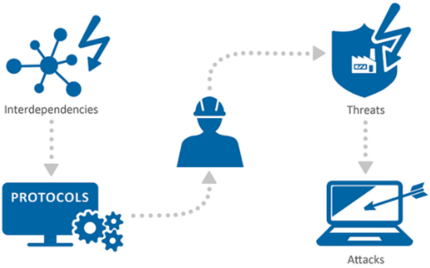


# Generation and Detection of Attacks in ICS (2/3)

- **OT is separated from the IT by a demilitarized zone (DMZ)**
  - No attacks from the IT systems can be launched against OT
- **Current trend in OT systems**
  - Use of embedded devices with computing capabilities
  - Use of Ethernet-based protocols
- **Increasing interconnection of OT systems with public networks**
  - E.g., Internet, mobile communications
- **Problem: New attack surfaces against OT systems**
  - Communication protocols support (almost) no security functions
  - (Field) devices are placed outside and are easily accessible to the attacker (e.g., via a wireless connection)

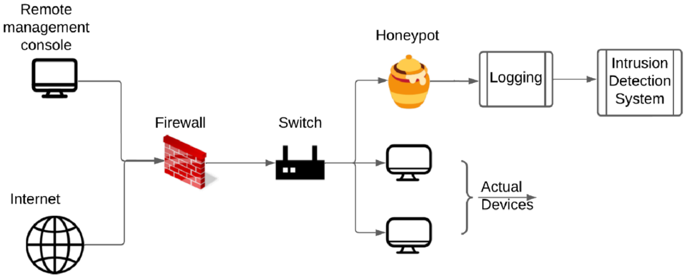
# Generation and Detection of Attacks in ICS (3/3)

- **Possible projects**
  1. Explore *different features* and machine learning methods for *detection of ICS attacks* from *network traffic*
  2. Investigate *different sets of features* extracted from *process data* and different machine learning algorithms in order to *detect attacks in ICS*
  3. Use of different *machine learning* algorithms to *extend the number of attacks* in existing ICS datasets



# Implementation and Evaluation of Honeypots

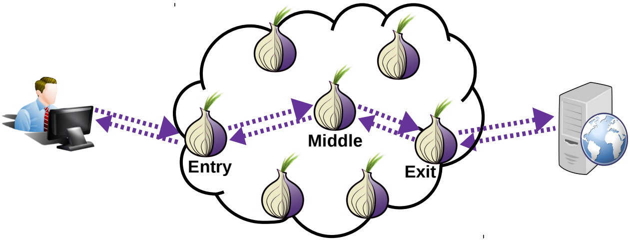
- **Goal of honeypots:** Decoys to deceive attackers
  - Attract real attackers
  - Record interactions useful for machine learning based attack detection
  - *Advantage:* Early warning about unknown attacks




- **Possible projects**
  - Implementation and evaluation of different types of honeypots, e.g.,
    1. Using deep neural networks or generative AI
    2. Using open-source network emulators

# End-to-end Traffic Correlation (1/2)

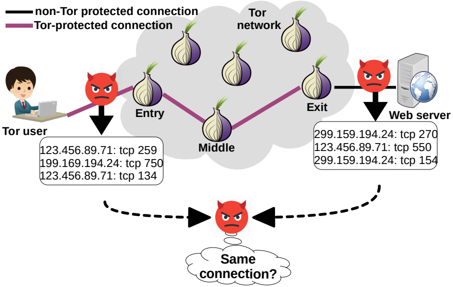
- **Privacy has become a concern**
  - People leave a large quantity of digital traces on the Internet
- **Encryption cannot fully obscure the content**
- **Solution: Privacy enchancing techniques**, e.g., Tor, VPN
  - Hide who communicates with whom




- *Problem:* **Traffic patterns are still visible**
  - Meta-data such as packet size, direction, and timestamps

# End-to-end Traffic Correlation (2/2)

- **Goal: Correlate ingress and egress segments of Tor connection**
  - Passive observation of traffic patterns
  - No need to break the encryption




- **Possible projects**
  1. Investigate the scalability of different *end-to-end traffic correlation attacks* in different evaluation scenarios
     - Use of sophisticated simulator for the generation of data
  2. Eventually, implementation and evaluation of different *defenses* against end-to-end traffic correlation

# Next Steps

- **Announcement of the list of selected students**
  - If you do not get place, do not deregister immediately
  - Let us know if you decide to drop the course (see Slide 4)
- **Creation of groups**
  - On volunteer basis in Moodle by Sunday
  - Please consider the restrictions, presented in Slide 3
  - If you cannot find group mate, we will select one for you
- **Topic selection**
  - Let us know your topic preferences in Moodle on Monday and Tuesday
  - *However*, final topic assignment is done by us

**Questions?**